PRACTICE SHEET

1. In a triangle ABC, $a = 2b$ and $\angle A = 3\angle B$. Which one of the following is correct?
(a) The triangle is isosceles
(b) The triangle is equilateral
(c) The triangle is right – angled
(d) Such triangle does not exist

2. The median AD of a triangle ABC is bisected at F, and BF is produced to meet the side AC in P. If $AP = \lambda AC$, then what is the value of λ?
(a) $\frac{1}{4}$ (b) $\frac{1}{2}$
(c) $\frac{2}{3}$ (d) $\frac{1}{3}$

3. If the perimeter of a triangle ABC is 30cm, then what is the value of $a\cos^2(C/2) + c\cos^2(A/2)$?
(a) 15cm (b) 10cm
(c) $\frac{15}{2}$ cm (d) 13cm

4. In ΔABC, if $\angle A : \angle B : \angle C = 1 : 2 : 3$, then what is BC : CA: AB?
(a) 1 : 2 : 3 (b) $1 : \sqrt{3} : 2$
(c) $2 : \sqrt{3} : 1$ (d) $\sqrt{3} : 1 : 2$

5. The angles A, B, C of a triangle are in the ratio 2 : 5 : 5. What is the value of tan B tan C?
(c) The triangle is right – angled
(d) The triangle is isosceles but not obtuse – angled

7. If median of the ΔABC through A is perpendicular to BC, then which one of the following is correct?
(a) $\tan A + \tan B = 0$
(b) $\tan B - \tan C = 0$
(c) $\tan C + 2\tan A = 0$
(d) $\tan B + \tan C = 0$

8. In a triangle ABC, $b = \sqrt{3}$ cm, $c = 1$ cm, $\angle A = 30°$, what is the value of a?
(a) $\sqrt{2}$ (b) 2 cm
(c) 1 cm (d) $\frac{1}{2}$cm

9. ABC is a triangle in which AB = 6m, BC = 8 cm and CA = 10 cm. What is the value of cot (A/4)?
(a) $\sqrt{5} - 2$ (b) $\sqrt{5} + 2$
(c) $\sqrt{3} - 1$ (d) $\sqrt{3} + 1$

10. If the sides of a triangle are 6cm, 10cm and 14cm, then what is the largest angle included by the sides?
(a) 90° (b) 120°
(c) 135° (d) 150°

11. In a triangle ABC, AC = 5 cm, AB = 7 cm, $BC = \sqrt{3}$. What is the measure of angle A ?
(a) 45^0 (b) 60^0